2.4 Solving Multi-Step Inequalities

Name _____ Period ____ Date _____

Algebra I 2.4 (30-minute plan) · Student sheet

I can…

- solve multi-step inequalities.
- identify inequalities with no solution or all real numbers as solutions.
- write and solve multi-step inequalities for real-life problems.

Vocabulary · fill in the missing word

Multi-step inequality — an inequality that _____ more than one step to solve

All real numbers — the _____ cancel and a true statement is left

No solution — the _____ cancel and a false statement is left

Part 1 · Notes and examples (fill in each note, then work the example)

Reversing the symbol: Reverse the symbol only when you multiply or divide by a _____.

Example 1

Solve $3x - 7 \geq 8$.

All real numbers: Variables cancel, true statement: _____.

No solution: Variables cancel, false statement: _____.

Example 2

Solve each inequality.

a. $4x + 2 > 4(x - 1)$ b. $3(x + 1) < 3x - 5$

Part 2 · Partner practice

Partner practice

Solve $-4x + 9 \leq 1$.